

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 02 October 2026

Team ID: PNT2022TMIDxxxxxx

Project Name: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Maximum Marks: 4 Marks

Technical Architecture & Technology Stack

Table 1: Technology Stack Components

S.No	Component	Description	Technology
1	User Interface	Service Portal interface where users view catalog items, configure laptop order choices, and submit requests.	Service Portal UI (HTML5, CSS3, JavaScript, AngularJS / ServiceNow Widgets)
2	Application Logic-1	Service Catalog Form logic, mandatory field validation, and approval workflow triggering.	ServiceNow Business Rules & Client Scripts (JavaScript)
3	Application Logic-2	Automated Catalog Task creation, dynamic short description updates, and assignment group routing.	ServiceNow Flow Designer (No-Code / Low-Code Workflow Engine)
4	Application Logic-3	Automated Email / Instance Notifications sent to approvers and requesters post-approval.	ServiceNow Notification Engine
5	Database	Relational relational data store for Service Requests (sc_request), Request Items (sc_req_item), and Catalog Tasks (sc_task).	ServiceNow Managed Database (MariaDB / Relational Architecture)

S.No	Component	Description	Technology
6	Cloud Database	Cloud-hosted database infrastructure storing user preferences, system logs, and execution state.	ServiceNow Enterprise Cloud PaaS
7	File Storage	Storage for attached approval documents, receipt attachments, and catalog item images.	ServiceNow Database Attachment Storage / Cloud Blob Storage
8	External API-1	REST API integration for pulling employee user details and reporting metrics.	ServiceNow Table API / REST API
9	External API-2	Integration with external identity management for single sign-on authentication.	SAML 2.0 / OAuth 2.0 API
10	Machine Learning Model	Virtual Agent / Predictive Intelligence model for routing and categorizing procurement requests.	ServiceNow Predictive Intelligence (Classification Model)
11	Infrastructure (Server / Cloud)	Cloud-native enterprise PaaS deployment hosting the ServiceNow developer instance.	Cloud Server Configuration: Multi-tenant Cloud Platform (ServiceNow Cloud Infrastructure)

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Standard frontend frameworks used for widget rendering in Service Portal.	AngularJS, Bootstrap, FontAwesome
2	Security Implementations	Role-Based Access Control (RBAC), ACLs for table restrictions, and encrypted	ServiceNow Access Control Lists (ACLs), TLS 1.3 Encryption,

S.No	Characteristics	Description	Technology
		web session management.	SAML 2.0 SSO, IAM Controls
3	Scalable Architecture	High-availability cloud architecture supporting simultaneous workflow processing across departments.	ServiceNow Enterprise Multi-Instance Architecture / Cloud Microservices
4	Availability	High uptime and redundancy provided by automated failover across data center pairs.	ServiceNow Advanced High Availability (AHA) Data Centers & Load Balancers
5	Performance	Asynchronous flow execution ensuring task creation finishes under 3 seconds under concurrent request loads.	Asynchronous Engine Processing, Edge Encryption & CDN caching